

Module 4: Rational Functions – Prequiz

Name:

Date:

Four short questions. Distinguish canceled factors from denominator factors that remain.

1. Divide $x^2 + 5x + 6$ by $x + 2$.

2. For

$$f(x) = \frac{x - 4}{(x - 1)(x + 2)},$$

find the vertical asymptotes and the x -intercept.

3. For

$$g(x) = \frac{(x + 3)(x - 1)}{(x + 3)(x + 2)},$$

find the hole as an ordered pair.

4. A rational function has numerator degree 1 and denominator degree 2. What is its horizontal asymptote?

Module 4: Rational Functions – Weekly Check-In

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Three questions. Distinguish canceled factors from denominator factors that remain.

1. Polynomial division.

Divide $2x^2 + 3x - 5$ by $x - 2$. Show the long division and then rewrite

$$R(x) = \frac{2x^2 + 3x - 5}{x - 2}$$

in quotient-remainder form.

2. A hole and asymptotes.

Consider

$$f(x) = \frac{(x+2)(x-4)}{(x+2)(x-1)(x+3)}.$$

Find the domain restrictions, the hole as an ordered pair, the vertical asymptotes, and the x -intercept.

Module 4: Rational Functions – Weekly Check-In, continued

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3. Intercepts and horizontal behavior.

Consider

$$g(x) = \frac{2x - 6}{x^2 + x - 6}.$$

- (a) Factor completely and state the domain restrictions.

- (b) Find any hole, vertical asymptote, x -intercept, and y -intercept.

- (c) Find the horizontal asymptote and explain how the numerator and denominator degrees determine it.